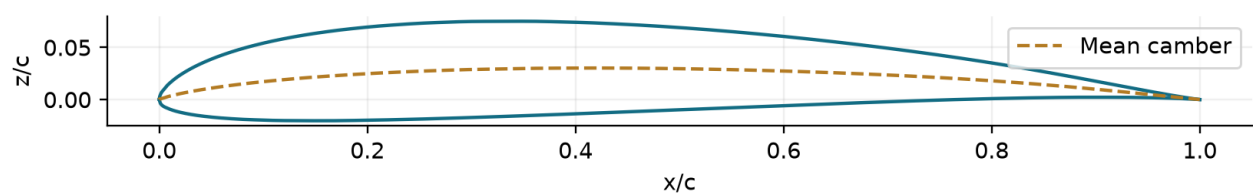


Airfoil & Aerodynamic Analysis

SD7037 selection, geometric derivations, trim and performance | GLIDER R03

Purpose and design status

This report documents the current R03 balsa glider geometry and its analytical aerodynamic model. It provides traceable inputs, numerical substitutions and coefficient definitions for engineering review. It is an English technical revision of the Persian report, not a flight-test report or a construction release.



Design quantity	Value
Wingspan / wing area	1.700 m / 0.255 m²
Root / tip chord	170 / 130 mm
Airfoil / estimated mass	SD7037 / 434.6 g
Reference speed	7.50 m/s
Predicted C_L / C_D	0.4851 / 0.03932
Predicted L/D at reference speed	12.34

Evidence level: wind-tunnel data are used for the two-dimensional section. The complete aircraft results are numerical predictions with assumed tail effectiveness, downwash and parasitic drag. No physical prototype test, CFD validation or dynamic stability assessment is claimed.

Contents: selection and geometry (pp. 2-3); notation and operating conditions (p. 4); section data (p. 5); lifting-line model (p. 6); trim (p. 7); drag (p. 8); stability (p. 9); performance (p. 10); technical review and sources (pp. 11-12).

01 / Airfoil selection

Decision basis and geometric section properties

SD7037 was selected as a practical candidate for this low-Reynolds-number glider. The available measured data bracket the reference operating Reynolds numbers; the section also provides finite thickness for a balsa spar and forward skin structure. This is a reasoned selection, not the outcome of a comparative airfoil optimization. [1, 2]

The coordinate file identifies SD7037-092-88. Upper and lower surfaces are interpolated with PCHIP, following the CAD generator; (0, 0) is added to close the leading edge. The values below are obtained from 10,001 chordwise samples. They are vertical surface separation and mean-line camber, not skin thickness.

$$t(x)/c = z_u(x/c) - z_l(x/c)$$
$$f(x)/c = [z_u(x/c) + z_l(x/c)] / 2$$

Derived quantity	Result
Maximum thickness / location	9.218% c / x/c = 0.2858
Maximum camber / location	3.022% c / x/c = 0.4153
Maximum root thickness	170 × 0.092184 = 15.671 mm
Maximum tip thickness	130 × 0.092184 = 11.984 mm

Trade-offs

Camber supports positive lift at modest angles of attack, but the negative section pitching moment requires balancing tail load. At the reference condition the tail carries a small downward force, increasing wing lift demand. A smooth, correctly shaped leading edge and accurate covering remain important; the clean wind-tunnel specimen is not identical to a built-up wooden wing.

The “D” in SD7037 (D) is the tested specimen identifier. The report does not imply a new airfoil designation or that the CAD surface exactly reproduces the tested specimen. The tail is modeled analytically; SD7037 data are not assigned to the tail.

02 / Planform and mass properties

Chosen inputs are distinguished from derived geometry

The planform dimensions were chosen within the supplied design ranges, then checked analytically. They are not uniquely determined by the aerodynamic equations. The wing uses zero leading-edge sweep, no geometric washout, 4° dihedral per side and 2° incidence.

$$\lambda = c_t/c_r = 130/170 = 0.764706$$
$$c(y) = 170 - 40|y|/850 \text{ mm, with } |y| \leq 850 \text{ mm}$$

$$S = b(c_r + c_t)/2 = 1.7(0.17 + 0.13)/2 = 0.255 \text{ m}^2$$
$$AR = b^2/S = 1.7^2/0.255 = 11.3333$$

$$\bar{c} = (2c_r/3)(1 + \lambda + \lambda^2)/(1 + \lambda) = 150.889 \text{ mm}$$
$$y_{MAC} = (b/6)(1 + 2\lambda)/(1 + \lambda) = 406.111 \text{ mm}$$

$$S_h = 0.4(0.105 + 0.090)/2 = 0.039 \text{ m}^2$$
$$AR_h = 0.4^2/0.039 = 4.10256; S_v = 0.01995 \text{ m}^2$$

$$X_{ac,w} = 220 + 0.25\bar{c} \cos(2^\circ) = 257.699 \text{ mm}$$
$$X_{CG} = 220 + 0.30\bar{c} \cos(2^\circ) = 265.239 \text{ mm}$$

$$l_{ac} = 540.000 \text{ mm}; l_{h,CG} = 532.460 \text{ mm}$$
$$V_h = S_h l_{ac}/(S\bar{c}) = 0.547345$$

$$m = \sum m_i = 434.615 \text{ g}; X_{CG} = \sum m_i X_i / \sum m_i$$
$$m/S = 434.615/25.5 = 17.0437 \text{ g/dm}^2$$

Mass includes 87.033 g of computed ballast, 17.267 g covering allowance and 22 g glue/finish/wiring allowance. These are design estimates. Reweigh the assembled aircraft and rebalance with the actual equipment. Wing area is the nominal planform area used throughout the model; exact dihedral/incidence projection corrections are neglected.

03 / Coordinates and definitions

Reference quantities, signs and operating point

X increases aft from the nose, Y runs spanwise and Z points upward. For local section coordinates u and v in mm, the CAD transformation is:

$$X = 220 + u \cos(2^\circ) + v \sin(2^\circ); Y = y$$
$$Z = 27 + |y| \tan(4^\circ) - u \sin(2^\circ) + v \cos(2^\circ)$$

Symbol	Meaning / convention
c_l, c_d, c_m	Two-dimensional section coefficients; reference q c and q c ² per unit span
C_L, C_D, C_m	Aircraft/wing coefficients; reference qS and $qS\bar{c}$ unless marked h
$C_{L,h}$	Tail coefficient based on tail area and local tail dynamic pressure
α, i, δ_e	Angle of attack, incidence and elevator deflection; positive δ_e is trailing-edge down
C_m sign	Positive is nose-up; derivatives are per radian
$\rho / \mu / g$	1.225 kg/m ³ / 1.789 × 10 ⁻⁵ Pa·s / 9.80665 m/s ²

$$q = \frac{1}{2}\rho V^2 = \frac{1}{2}(1.225)(7.5^2) = 34.4531 \text{ Pa}$$
$$W = mg = 4.26212 \text{ N}; C_{L,req} = W/(qS) = 0.485129$$

$$Re = \rho V c / \mu$$
$$Re_{root} = 87,304; Re_{tip} = 66,762; Re_{MAC} = 77,490$$

Air properties are assumed, not measured in the hall. The baseline solver uses $L \approx W$. In exact steady glide, $L = W \cos \gamma$; at the reference glide angle this approximation is about 0.33%. The tail dynamic-pressure ratio $\eta_h = 0.90$ is assumed.

04 / Experimental section data

Two-stage interpolation with separate lift/moment and drag Reynolds grids

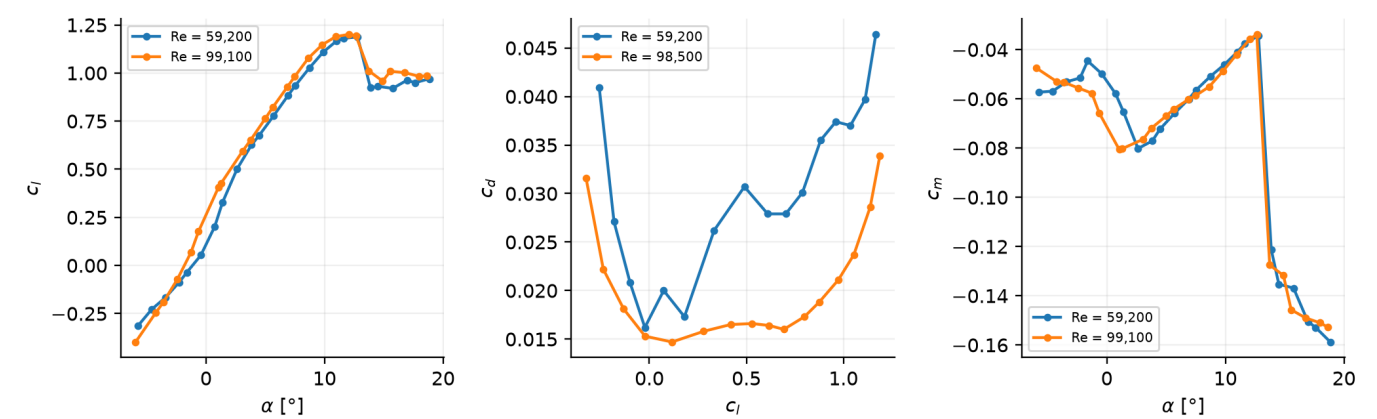
The analysis uses SD7037D.LFT and SD7037D.DRG, labeled “clean”. The ascending-angle branch is retained up to the first angle reversal. Values are linearly interpolated first in α and then in Re . The lift/moment and drag runs have different Reynolds numbers. Data were produced under the UIUC Low-Speed Airfoil Test program. [2]

Dataset	Measured Reynolds numbers
Lift and moment	59,200; 99,100; 197,400; 296,000
Drag	59,200; 98,500; 196,600; 295,100

$$F(\alpha) = F_1 + (F_2 - F_1)(\alpha - \alpha_1)/(\alpha_2 - \alpha_1)$$
$$F(Re) = F_{lo} + (F_{hi} - F_{lo})(Re - Re_{lo})/(Re_{hi} - Re_{lo})$$

Worked example: $\alpha = 3^\circ$ and $Re = 80,000$. The Reynolds interpolation weights are 0.521303 for lift/moment and 0.529262 for drag. This is a section example, not the aircraft operating coefficient.

Coefficient	Low-Re value	High-Re value	At $Re = 80,000$
c_l	0.539917	0.585622	0.563743
c_m	-0.079252	-0.076573	-0.077856
c_d	0.028409	0.016455	0.022082



Endpoint clamping in the interpolation routine is not physical extrapolation. The review checks both datasets against the bracketing Re/α domains rather than relying only on the original low-Re flag. The wind-tunnel curves are not a whole-aircraft stall prediction.

05 / Finite-wing solution

Nonlinear lifting-line model and numerical discretization

Fourteen odd Fourier terms are used for the symmetric wing. Unknowns are 14 circulation coefficients, body angle of attack and tail lift coefficient. Fourteen section-matching equations plus lift and pitching-moment equilibrium form a 16-equation nonlinear system.

$$y = (b/2) \cos \theta; n = 1, 3, \dots, 27$$
$$\Gamma(\theta) = 2bV \sum A_n \sin(n\theta)$$

$$c_l(y) = (4b/c(y)) \sum A_n \sin(n\theta)$$
$$\alpha_i = \sum n A_n \sin(n\theta) / \sin \theta; \alpha_{\text{eff}} = \alpha_{\text{body}} + i_w - \alpha_i$$

$$C_{L,w} = \pi AR A_1$$
$$C_{D,i,w} = \pi AR \sum n A_n^2$$

$$C_{D,p,w} = (2/S) \int_0^{b/2} c_d(y) c(y) dy$$
$$C_{m,w} = [2/(Sc)] \int_0^{b/2} c_m(y) c(y)^2 dy$$

$$A_1 = 0.01404074; C_{L,w} = 0.499917$$
$$C_{D,i,w} = 0.00731910; C_{m,w} = -0.0771794$$

Spanwise integrations use 301 points on the half-span. The maximum reference section lift coefficient is 0.53946. Angles within the equations are in radians, then converted to degrees for lookup. The solution is a planar lifting-line approximation; it does not resolve fuselage flow, all wing-tail interference or separated three-dimensional flow.

Quarter-chord is used as the section-moment reference in the model. The tabulated file header does not independently specify that reference; this assumption should be confirmed against the original test documentation before a higher-fidelity trim validation. No unverified reference is presented as a measured fact.

06 / Longitudinal trim

Simultaneous force balance and moment balance about the CG

$$C_{L,w} + \eta_h (S_h/S) C_{L,h} = W/(qS)$$
$$0.499917 + 0.9(0.039/0.255)(-0.107433) = 0.485129$$

$$C_{m,w} + C_{L,w}(h - 0.25) - \eta_h (S_h/S) C_{L,h}(l_{h,CG}/\bar{c}) = 0$$
$$-0.077179 + 0.024996 + 0.052184 \approx 0$$

Here $h = 0.30$ is the normalized CG coordinate. The wing term uses $h - 0.25$ rather than an exact inclined geometric lever arm; force directions are treated in the small-angle approximation. Baseline fuselage pitching moment is assumed zero. The tail moment arm is 532.460 mm, not the 540 mm aerodynamic-center spacing.

$$a_h = 2\pi AR_h / [2 + \sqrt{(4 + AR_h^2)}] = 3.92699 \text{ rad}^{-1}$$
$$\varepsilon = 2C_{L,w}/(\pi AR) = 1.6090^\circ$$

$$\delta_e = [C_{L,h}/a_h - (a_{\text{body}} + i_h - \varepsilon)]/\tau_e$$
$$i_h = 0^\circ; \tau_e = 0.55 \text{ (assumed)}$$

Reference trim output	Value
Body angle of attack	1.36565°
Elevator deflection	-2.40759° (trailing-edge up)
Tail lift	-0.129919 N
Wing lift	4.392040 N
Total lift	4.262121 N

The calculated deflection is inside the modeled $\pm 15^\circ$ elevator range. This is a predicted equilibrium setting; actual hinge gaps, linkage geometry and tail effectiveness require physical checking. A negative wing pitching moment is compatible with zero net aircraft moment at trim.

07 / Drag build-up

All aircraft drag contributions normalized by qS

$$C_D = C_{D,p,w} + C_{D,i,w} + C_{D,h} + C_{D,v} + C_{D,body} + C_{D,extra}$$

$$C_{D,h} = \eta_h(S_h/S)[0.012 + C_{L,h}^2/(\pi \times 0.8 \times AR_h)]$$
$$C_{D,v} = (S_v/S) \times 0.012$$

$$C_{D,body} = C_f FF S_{wet}/S$$
$$= 0.004 \times 1.3 \times 0.15 / 0.255 = 0.00305882$$

Contribution	C _D	Basis
Wing profile	0.0231934	Measured-section interpolation
Wing induced	0.0073191	Circulation solution
Horizontal tail	0.0018058	Analytical + assumed terms
Vertical tail	0.0009388	Assumed profile drag
Fuselage	0.0030588	Estimated wetted-area model
Additional allowance	0.0030000	Seams, roughness, hardware
Total	0.0393160	Model sum

$$D = qSC_D = 34.453125 \times 0.255 \times 0.03931598 = 0.345412 \text{ N}$$
$$L/D = C_L/C_D = 12.3392$$

The 0.012 tail/fin profile drag, tail efficiency 0.8, body friction 0.004, form factor 1.3, wetted area 0.15 m² and extra drag 0.003 are assumptions. Reporting many decimal places would not make these assumptions equally accurate. Values are retained here for traceability, not as an uncertainty claim.

08 / Static longitudinal stability

Local, fixed-elevator derivatives near the reference condition

The wing solution is repeated at body angle $\alpha \pm \Delta\alpha$ with $\Delta\alpha = 0.15^\circ$. A central difference estimates the local slopes. A destabilizing fuselage derivative of $+0.08 \text{ rad}^{-1}$ is included here as an assumption; it is not a calibrated fuselage model.

$$C_{L\alpha,w} \approx [C_{L,w}(\alpha + \Delta\alpha) - C_{L,w}(\alpha - \Delta\alpha)]/(2\Delta\alpha)$$
$$\varepsilon_\alpha = 2C_{L\alpha,w}/(\pi AR) = 0.290211$$

$$C_{L\alpha,h} = a_h(1 - \varepsilon_\alpha)$$
$$C_{L\alpha} = C_{L\alpha,w} + \eta_h(S_h/S)C_{L\alpha,h}$$

$$C_{ma} = C_{ma,w} + (h - 0.25)C_{L\alpha,w}$$
$$- \eta_h(S_h/S)(l_{h,CG}/\bar{c})C_{L\alpha,h} + 0.08$$

$$C_{L\alpha} = 5.55011 \text{ rad}^{-1}; C_{ma} = -1.08774 \text{ rad}^{-1}$$
$$SM = -C_{ma}/C_{L\alpha} = 0.195986 = 19.60\% \text{ MAC}$$

$$X_{NP} \approx X_{CG} + SM \bar{c} = 294.811 \text{ mm}$$

The negative moment slope indicates local static longitudinal stability under the model assumptions. The neutral-point conversion follows the baseline code and neglects the incidence cosine. It does not establish dynamic-mode damping, lateral/directional stability or post-stall behavior.

Numerical convergence is not physical validation

Quantity at 7.5 m/s	14 terms	20 terms
C_D	0.03931598	0.03931560
L/D	12.339225	12.339345
Elevator trim	-2.40759°	-2.40886°
Static margin	19.5986%	19.6621%

These changes quantify sensitivity to Fourier resolution only. They do not cover data interpolation, uncertainty in tail/body terms, build accuracy or measured flight performance.

09 / Performance envelope

Best sampled point within the checked data domain

$$L/D = C_L/C_D; \gamma = \text{atan}[1/(L/D)]$$
$$V_{\text{sink}} = V \sin \gamma; R/H = \cot \gamma = L/D$$

Quantity	7.50 m/s reference	6.75 m/s sampled best
C_L	0.485129	0.598924
C_D	0.039316	0.044622
L/D	12.339	13.422
Sink rate	0.60583 m/s	0.50151 m/s
Elevator trim	-2.408°	-3.910°

The speed grid is 6.50-10.50 m/s in steps of 0.25 m/s. The 6.50 m/s point is excluded because the tip Reynolds number falls below measured coverage. The new review also checks angle/Re bounds in both input datasets at the 301 integration stations. The selected 6.75 m/s point is near the lower data boundary; it is not a continuous optimum or a stall-speed determination.

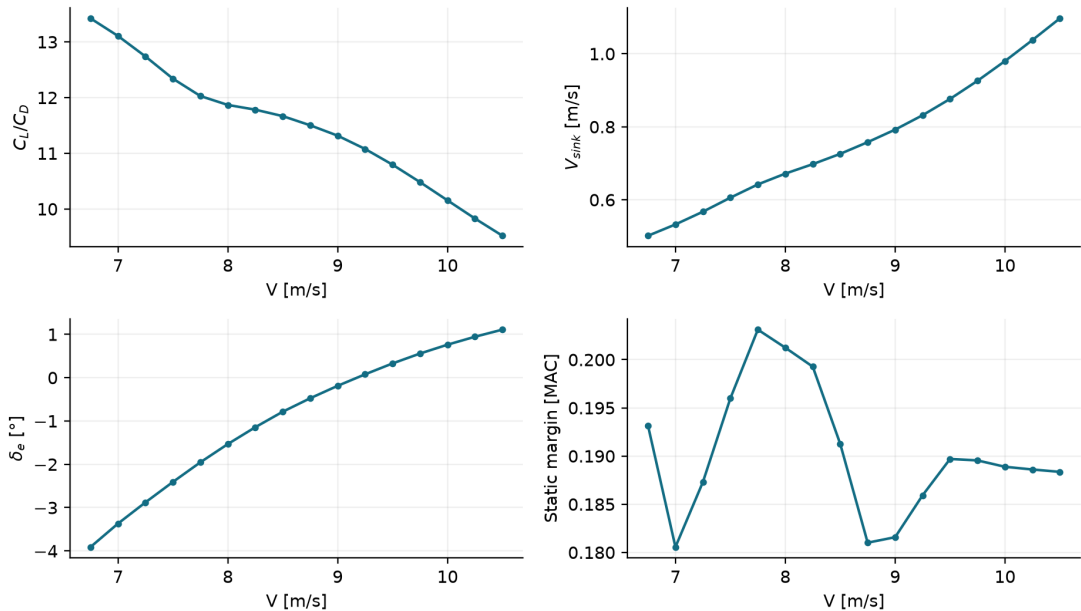


Illustration only: a stabilized, straight glide through a 2 m height loss gives $2 \times 13.422 = 26.84$ m in still air. This excludes launch transients, turns and recovery. It is not a prediction of competition-course distance without the actual launch and course conditions.

10 / Technical review findings

Corrections and remaining limitations in this English edition

Notation and numerical checks

Greek symbols, subscripts and exponents are used consistently. Lowercase section coefficients are distinguished from finite-wing/aircraft coefficients. Δa denotes the finite-difference step; da denotes a differential. The section moment example uses c_m . The reference solve was rerun and matches the archived C_L , C_D , L/D and static margin to 10^{-9} or better numerically.

Endpoint treatment in the baseline solver

At the exact wing tip, the baseline implementation regularizes $\sin \theta$ with a small denominator. Because the numerator is exactly zero there, it does not evaluate the analytical limiting downwash. The mathematical limit is $\alpha_{i,tip} = \sum n^2 A_n$. This affects the endpoint used in the trapezoidal drag/moment integral.

At 7.5 m/s the baseline endpoint α_{eff} is 3.3657° , whereas the limiting value is -0.5434° . Replacing only that endpoint, with circulation and trim held fixed, changes wing profile C_D by -0.0000107 . This is a diagnostic sensitivity, not a fully retrimmed correction. Baseline results remain explicitly labeled; no model change is hidden in the translation.

Source and validity qualifications

The locally bundled readme.TXT is labeled Volume 1 and should not be used as proof of the volume provenance of SD7037D. This edition cites the actual LFT/DRG file identifiers and the official UIUC data index. The quarter-chord moment reference remains a documented modeling assumption pending verification in the original test report.

No reliable whole-aircraft $C_{L,max}$, stall speed, dynamic derivatives or lateral/directional stability derivatives have been established. Airfoil surface condition, covering deformation, tail effectiveness and actual mass properties remain important sources of uncertainty.

Example drag sensitivity at fixed reference lift:
 $C_L / (C_D + 0.005) = 0.485129 / 0.044316 = 10.947$

Before flight correlation, measure actual mass, CG, incidence, air conditions and control travel. Compare measured speed and sink rate with the model. This report does not certify structural strength, flutter clearance or competition-rule compliance.

11 / References and traceability

Reviewed 04 October 2026 | No personal workstation paths included

[1] UIUC Airfoil Coordinates Database. sd7037.dat, header SD7037-092-88.
m-selig.ae.illinois.edu/ads/coord/sd7037.dat

[2] UIUC Low-Speed Airfoil Tests. Experimental files SD7037D.LFT and SD7037D.DRG, clean specimen, builder listed as H. Stokely / B. Williams. Official data and documentation index:
m-selig.ae.illinois.edu/pd.html
m-selig.ae.illinois.edu/uiuc_lsar.html

The experimental airfoil performance data were produced under the UIUC Low-Speed Airfoil Test program. They are third-party data, not newly measured results from this project. Consult the provider’s notices and license when redistributing raw datasets; coordinate-file terms should not be inferred from the polar-data license.

Project calculation records

File / record	Role
source/build_structure.py	Geometry and mass generation
source/analyze_glider.py	Nonlinear lifting-line, trim and beam screening
source/check_aero_convergence.py	14-versus-20-term sensitivity
analysis/geometry.json; mass_balance.json	Final geometry and estimated mass inputs
analysis/engineering_results.json	Archived baseline numerical outputs
source/build_aero_report_en.py	English report and independent domain audit

Analysis paths above are relative to output/Glider_R03 where applicable. Reproduction requires the local Python dependencies plus the original coordinate/polar files. The report generator evaluates the aerodynamic functions without rerunning the CAD build. Source hashes below identify the reviewed calculation inputs.

```
analyze_glider.py (SHA-256)
e15338f105f1127a93daf188d1a0a4d3c3be6a71d4a7701de357bc4b38e46b65

sd7037.dat (SHA-256)
58dc2b375e8c96c4b2ad7acd2968de9d69334ec80c656da10a022981e470784f

SD7037D.LFT (SHA-256)
6aa017eff3759b1e195ec190411f82a8b97b83e94a6af4c85784a4636b2ca17a

SD7037D.DRG (SHA-256)
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